Active Learning for Employee Attrition

Team: <Name 1> · <Name 2> · <Name 3>

Pool ~14,900 unlabeled $\longrightarrow$ Oracle ($\leq$5,000 queries) $\longrightarrow$ fixed Random Forest

Metric	F1 on the minority *Left* class, fixed 0.5 threshold, mean over 3 seeds
Free start	500 labeled samples per seed
Limits	5,000 unique oracle queries · 60 s runtime per seed · RF only

Our Pipeline: Margin Sampling + Positive Hunt + Oversampling

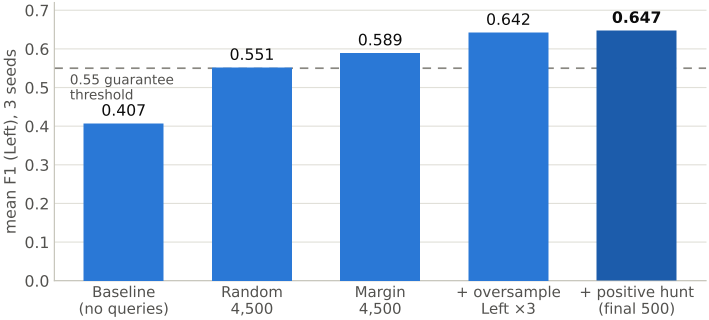
1. **Margin sampling** — 9 rounds $\times$ 500: retrain, then query the pool samples with smallest $|P(\mathrm{Left}) - 0.5|$ 4,500 queries

2. **Positive hunt** — last 500 queries go to the *highest* $P(\mathrm{Left})$: real minority samples, not more boundary 500 queries

3. **Minority oversampling** — duplicate every *Left* row $\times$3 before the final fit 0 queries

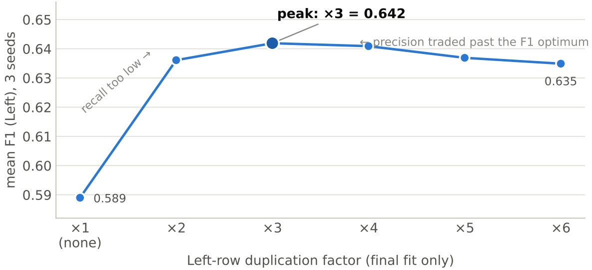
Oversampling is applied at the *final fit only* — inside the loop it would skew the probabilities that margin scoring depends on.

What Each Component Was Worth



Oversampling is the single biggest win (+0.053), for zero extra budget.

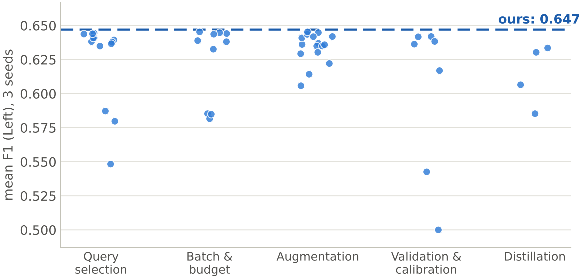
Why $\times 3$? The Oversampling Factor Is a Threshold Dial



RF hyperparameters and encoding are fixed by the framework — the training distribution is our only tuning lever.

Duplication shifts the fixed 0.5 threshold implicitly; F1 peaks at $\times 3$.

What Did Not Work — and Why



Key finding: after querying, the training set is ∼51% positive vs. ∼33% in the pool — so any hyperparameter validated on queried data does not transfer.

Results & Conclusion

	Seed 1	Seed 2	Seed 3	**Mean**
F1 (Left)	0.6383	0.6493	0.6535	**0.6470**

- Runtime 20–30 s of the 60 s budget per seed
- Top alternatives sit within seed-level noise ($\pm$0.005–0.01)
- Stopped after 3 passes — further local search would overfit the local test sets

Simple, budget-aware components beat every clever variant we tried.